CARBOHYDRATES 5

Carbohydrates are the most common class of organic (carbon-based) compounds present in organisms. Their name comes from *carbon hydrates* because many of carbohydrates have a formula that can be written as $C_x(H_2O)_y$. Many carbohydrates are formed by the process of **photosynthesis**. In photosynthesis, sunlight provides the energy to change carbon dioxide and water to carbohydrates in the presence of chlorophyll. The most common carbohydrate is **glucose** ($C_6H_{12}O_6$).

1 Copy and balance the following reaction to show the formation of glucose by photosynthesis.
CO_2 (g) + $H_2O \rightarrow C_6H_{12}O_6$

2 Is photosynthesis an *exothermic* reaction (heat given out) or *endothermic* reaction (heat taken in)? Explain your answer.

Glucose is a monosaccharide; it can't be broken down into two or more simpler sugars. Glucose is a hexose – there are 6 carbon atoms per molecule. Fructose is a similar molecule. A molecule of glucose can join with a molecule of fructose to form the disaccharide, sucrose. A molecule of water is also formed in this process. Figure 1 shows how glucose and fructose combine to form sucrose.

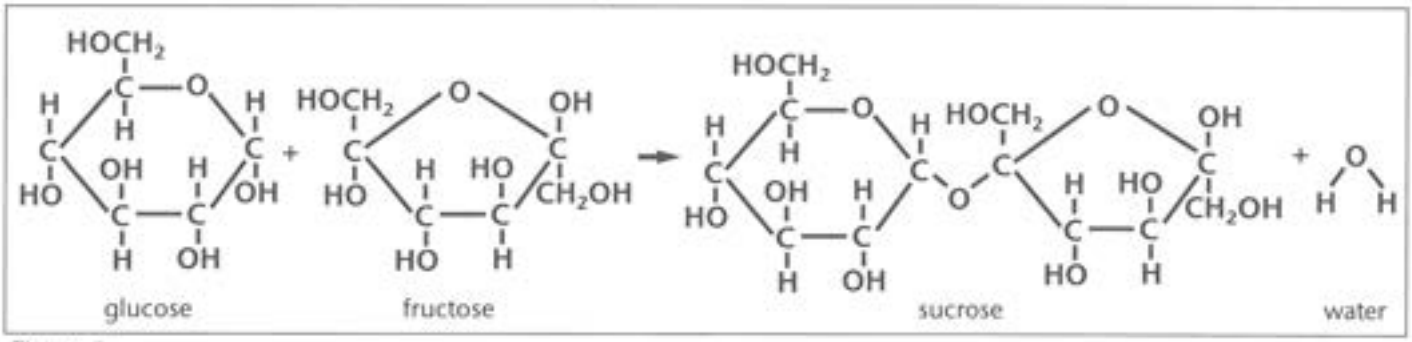


Figure 1

3 Copy figure 1.

4 Use figure 1 to help you write a balanced equation for the formation of sucrose.

5 Identify one difference between a glucose molecule and a fructose molecule.

6 Explain why sucrose is a called a *disaccharide*.

7 The formation of sucrose is a condensation reaction. What is meant by *condensation*?

8 What is a *hexose*?

Polysaccharides are formed by joining many monosaccharides together. Cellulose, starch and glycogen are all polysaccharides that use glucose as the monomer (basic 'building block'). Cellulose has the formula $(C_6H_{11}O_5)n$, where n can be from around 500 to 5000. Cellulose is being investigated as a replacement for petroleum hydrocarbons, because it is a renewable resource (being a major component of plants). Cellulose is not soluble in water. Starch is also derived from plants and insoluble in cold water but the glucose units are bonded in a different way. Approximately 3000 glucose units will make up a typical starch molecule. The test for starch is that it gives a blue/black colour when added to iodine solution. Glycogen (animal starch) has a more branched structure than starch and it is soluble in water. Glycogen is used by vertebrates for food storage.

9 List the similarities and differences between cellulose, starch and glycogen.

10 What is the chemical test for starch?

11 Why is cellulose being considered as a replacement for petroleum hydrocarbons?

Reducing sugars have a free group for chemical bonding. They react with copper (II) ions (which are blue) to copper (I) compounds (which are red). They are called 'reducing' sugars because they can reduce copper (II) to copper (I); i.e. Cu^{2+} to Cu^{+}. Benedict's solution is the most common reagent used. (A **reagent** is a substance used to measure or detect the presence of other substances.) Glucose, fructose, maltose (malt sugar) and lactose (milk sugar) are reducing sugars. Sucrose is not a reducing sugar.

12 Which ions in Benedict's solution are responsible for identifying reducing sugars?

© Peter Gribben, Wendy St Germain and Blake Education 2005
© Blake Education ISBN 978 1 92561 067 3 Assignments in Chemistry Book 3